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# EXPERIMENT 9

**Exp No:9 Experiment to understand Matplotlib library use cases in Data Science through visualization**

**AIM:**
To understand Matplotlib library use cases in Data Science through visualization

**DESCRIPTION:**
Use Titanic data set to understand the Matplotlib library.

**CODE:**

```
import matplotlib.pyplot as plt

%matplotlib inline

data = pd.read_csv('/content/Titanic-Dataset.csv')

pd.set_option('display.max_columns', None)

pd.set_option('display.width', None)

display(data)

data.info()

display(data.describe())

display(data.value_counts('Survived'))

sns.countplot(x='Survived', data=data)

plt.show()

dummies = pd.get_dummies(data.Sex)

FinalDataset = pd.concat( [pd.get_dummies(data.Sex), data.iloc[:,[0,1,2,5]]],axis=1)

display(FinalDataset.head())

sns.scatterplot(x='Age',y='Fare', hue='Survived',data=data)
```

```
plt.show()
sns.scatterplot( x='Pclass', y='Fare',hue='Survived',data=data)
plt.show()
sns.pairplot( data[['Survived', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare']],hue='Survived',
height=3)
plt.show()
sns.FacetGrid( data,hue='Survived', height=5).map(sns.histplot, 'Age').add_legend()
plt.show()
sns.FacetGrid( data, hue='Survived', height=5).map(sns.histplot, 'Fare').add_legend()
plt.show()
sns.FacetGrid(data,hue='Survived', height=5).map(sns.histplot, 'Pclass').add_legend()
plt.show()
sns.FacetGrid( data,hue='Survived', height=5).map(sns.histplot, 'SibSp').add_legend()
plt.show()
```

**OUTPUT:**

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 0 | 3 | Braund, Mr. Owen Harris | male | 22.0 | 1 | 0 | A/5 21171 | 7.2500 | NaN | S |
| **1** | 2 | 1 | 1 | Cumings, Mrs. John Bradl | female | 38.0 | 1 | 0 | PC 17599 | 71.2833 | C85 | C |

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | ey (Florence Briggs Th... | | | | | | | | |
| **2** | 3 | 1 | 3 | Heikkinen, Miss. Laina | female | 26.0 | 0 | 0 | STON/O2. 3101282 | 7.9250 | NaN | S |
| **3** | 4 | 1 | 1 | Futrelle, Mrs. Jacques Heath (Lily May Peel) | female | 35.0 | 1 | 0 | 113803 | 53.1000 | C123 | S |
| **4** | 5 | 0 | 3 | Allen, Mr. William Henry | male | 35.0 | 0 | 0 | 373450 | 8.0500 | NaN | S |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |

| PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Cabin | Embarked |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **886** | 887 | 0 | 2 | Montvila, Rev. Juozas | male | 27.0 | 0 | 0 | 211536 | 13.0000 | NaN | S |
| **887** | 888 | 1 | 1 | Graham, Miss. Margaret Edith | female | 19.0 | 0 | 0 | 112053 | 30.0000 | B42 | S |
| **888** | 889 | 0 | 3 | Johnston, Miss. Catherine Helen "Carrie" | female | NaN | 1 | 2 | W./C. 6607 | 23.4500 | NaN | S |
| **889** | 890 | 1 | 1 | Behr, Mr. Karl Howell | male | 26.0 | 0 | 0 | 111369 | 30.0000 | C148 | C |
| **890** | 891 | 0 | 3 | Dooley, Mr. Patrick | male | 32.0 | 0 | 0 | 370376 | 7.7500 | NaN | Q |

891 rows × 12 columns

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

| # | Column | Non-Null Count | Dtype |
|---|---|---|---|
| 0 | PassengerId | 891 non-null | int64 |
| 1 | Survived | 891 non-null | int64 |
| 2 | Pclass | 891 non-null | int64 |
| 3 | Name | 891 non-null | object |
| 4 | Sex | 891 non-null | object |
| 5 | Age | 714 non-null | float64 |
| 6 | SibSp | 891 non-null | int64 |
| 7 | Parch | 891 non-null | int64 |
| 8 | Ticket | 891 non-null | object |
| 9 | Fare | 891 non-null | float64 |
| 10 | Cabin | 204 non-null | object |
| 11 | Embarked | 889 non-null | object |

dtypes: float64(2), int64(5), object(5)

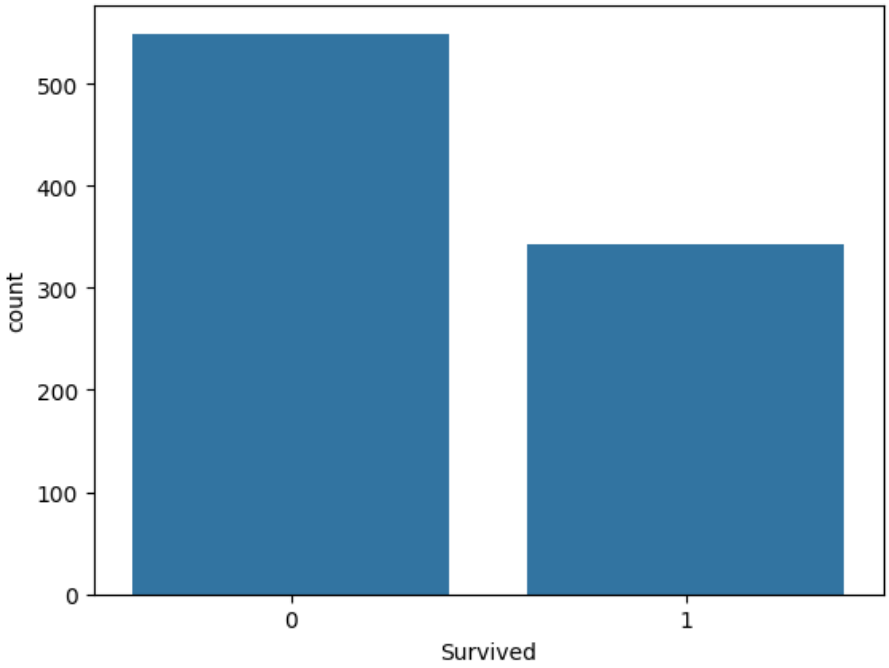
memory usage: 83.7+ KB

| | PassengerId | Survived | Pclass | Age | SibSp | Parch | Fare |
|---|---|---|---|---|---|---|---|
| **count** | 891.000000 | 891.000000 | 891.000000 | 714.000000 | 891.000000 | 891.000000 | 891.000000 |

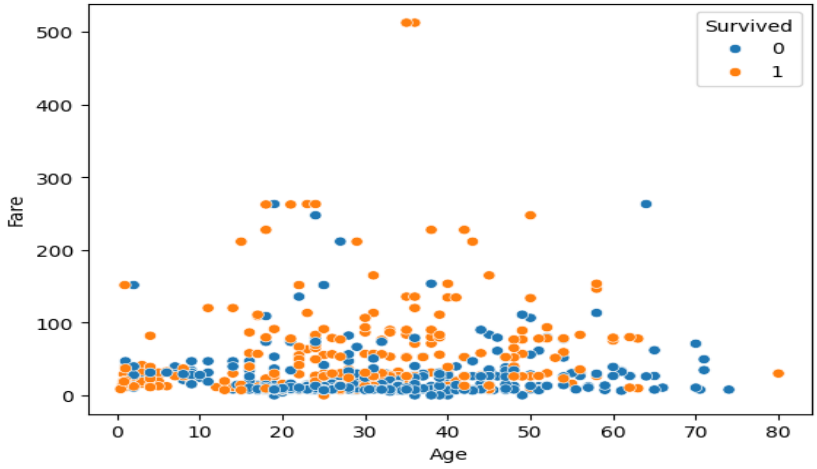
| | PassengerId | Survived | Pclass | Age | SibSp | Parch | Fare |
|---|---|---|---|---|---|---|---|
| **mean** | 446.000000 | 0.383838 | 2.308642 | 29.699118 | 0.523008 | 0.381594 | 32.204208 |
| **std** | 257.353842 | 0.486592 | 0.836071 | 14.526497 | 1.102743 | 0.806057 | 49.693429 |
| **min** | 1.000000 | 0.000000 | 1.000000 | 0.420000 | 0.000000 | 0.000000 | 0.000000 |
| **25%** | 223.500000 | 0.000000 | 2.000000 | 20.125000 | 0.000000 | 0.000000 | 7.910400 |
| **50%** | 446.000000 | 0.000000 | 3.000000 | 28.000000 | 0.000000 | 0.000000 | 14.454200 |
| **75%** | 668.500000 | 1.000000 | 3.000000 | 38.000000 | 1.000000 | 0.000000 | 31.000000 |
| **max** | 891.000000 | 1.000000 | 3.000000 | 80.000000 | 8.000000 | 6.000000 | 512.329200 |

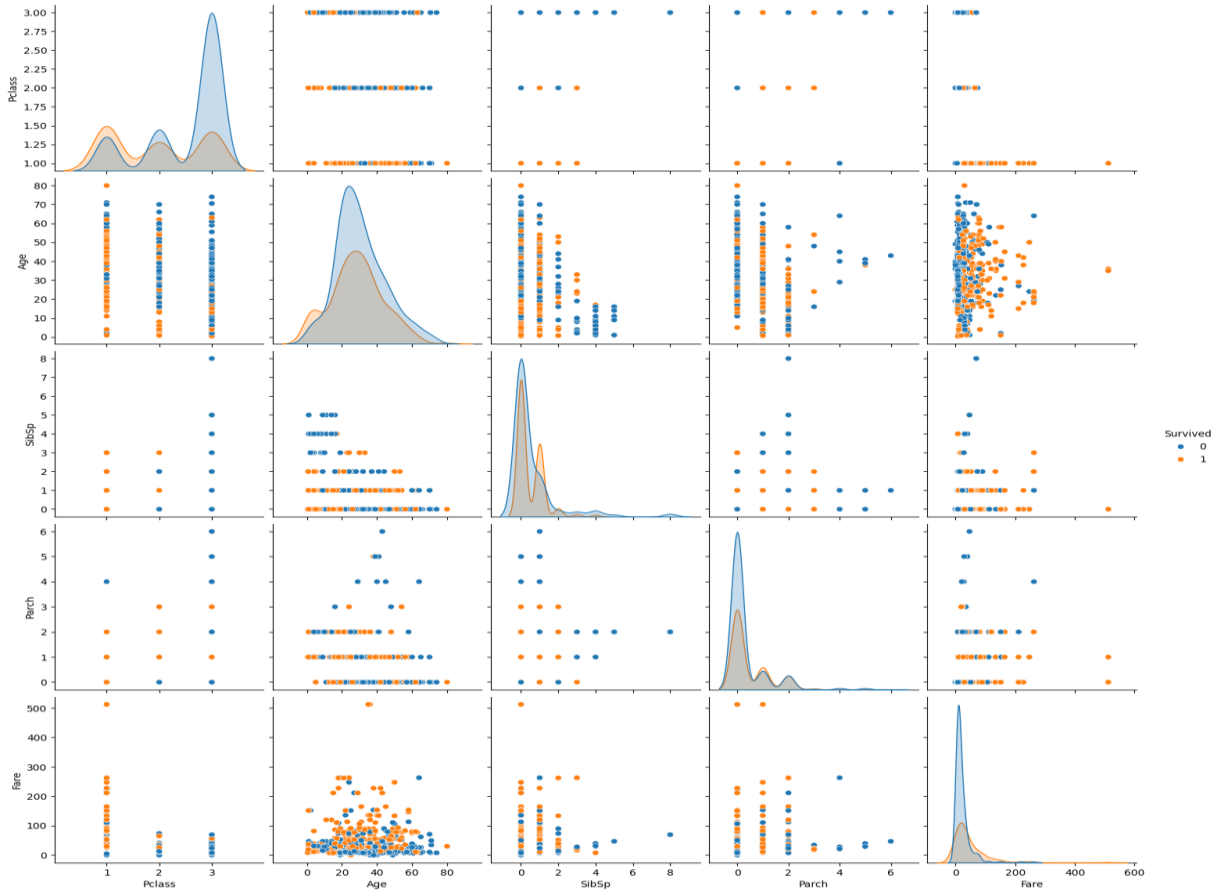
| **Survived** | **count** |
|---|---|
| **0** | 549 |
| **1** | 342 |

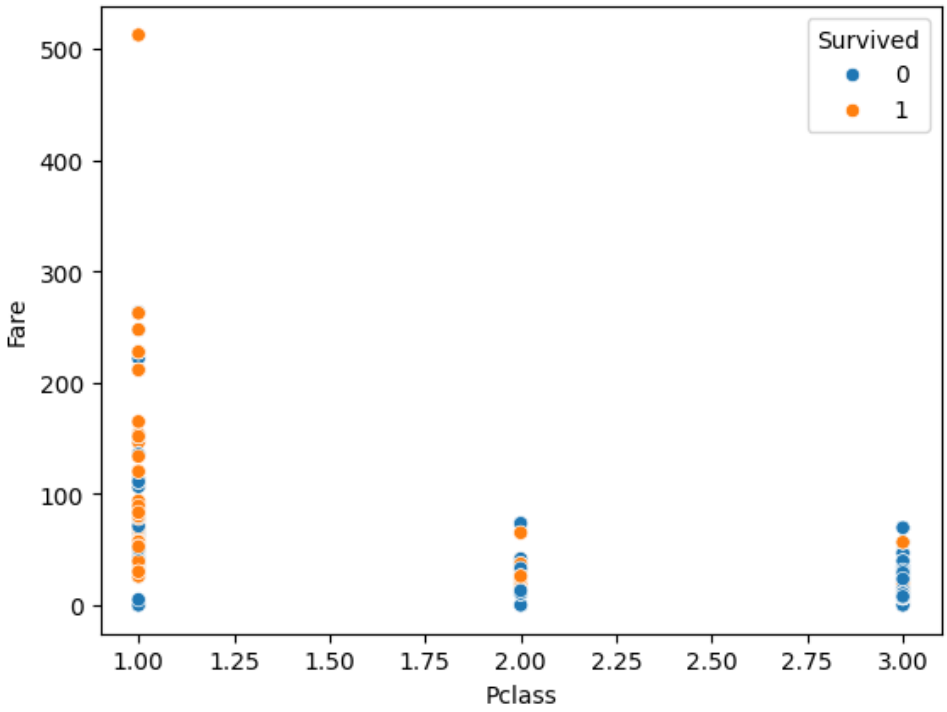
**dtype:** int64

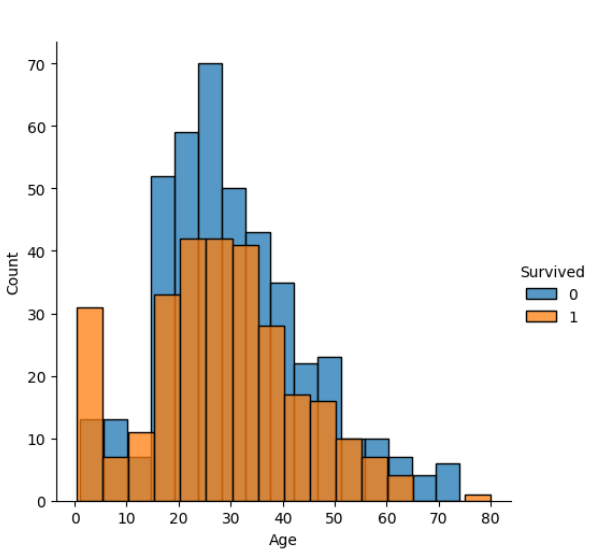


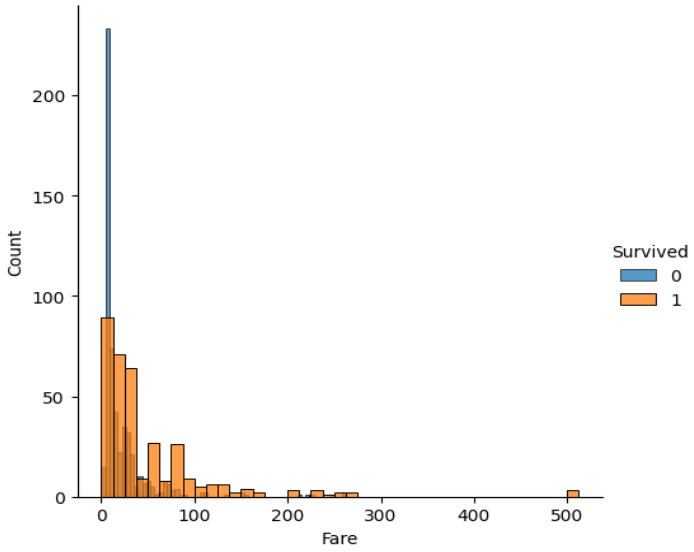

| | female | male | PassengerId | Survived | Pclass | Age |
|---|---|---|---|---|---|---|
| **0** | False | True | 1 | 0 | 3 | 22.0 |
| **1** | True | False | 2 | 1 | 1 | 38.0 |
| **2** | True | False | 3 | 1 | 3 | 26.0 |
| **3** | True | False | 4 | 1 | 1 | 35.0 |
| **4** | False | True | 5 | 0 | 3 | 35.0 |

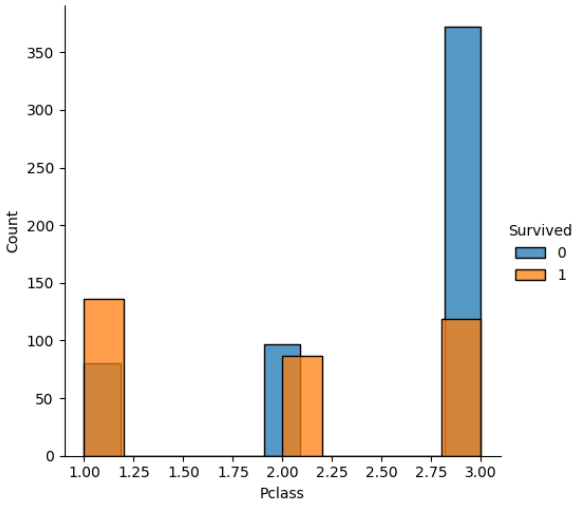


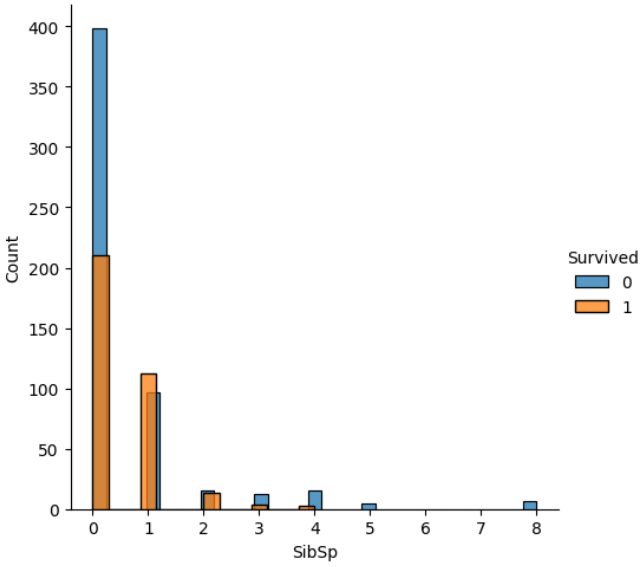

Pclass
Age
SibSp
Parch
Fare
Survived
0
1


Survived
0
1
Fare
Pclass




**RESULT:**

The Titanic dataset was successfully analyzed using Pandas, Matplotlib, and Seaborn with various statistical analyses and visualizations.